Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML, CSS, Bootstrap (Flask Jinja2 templates*	*Lightweight and simple to build a responsive, user-friendly interface for the Home, About, and FindYourCrop pages*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python 3, Flask*	*Flask is a lightweight WSGI framework, easy to integrate directly with the trained Scikit-learn model for fast, real-time crop predictions*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*CSV Dataset (Crop_recommendation.csv) + Pickle (model.pkl) file storage*	*Project uses a static agricultural dataset for training and a serialized model file for predictions, so a full relational/NoSQL database wasn't required*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Local Flask Development Server (can be deployed to Heroku / AWS in future)*	*Sufficient for development and testing; can scale to cloud hosting as usage grows*
5	**Version Control & CI/CD**	*github*	*Enables version tracking and collaboration on the OptiCrop codebase*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Kaggle (dataset source)*	*Source of the agricultural dataset (soil nutrients, climate parameters, crop labels) used to train the model*